

# ROHAN TEJA NALLAPANENI

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## Summary

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Full-stack Software Engineer and final-year Computer Science student (CGPA 8.64/10) with hands-on experience independently designing, building, and deploying six production-grade applications spanning AI/LLM orchestration, machine learning, and full-stack web development. Skilled in Java/Spring Boot backends, React/Next.js frontends, relational and graph/vector databases, and cloud deployment (AWS, Cloudflare Workers, Docker). Published IEEE-format research on ML-driven Android malware detection at an international conference. Strong foundation in Data Structures, Algorithms, System Design, and Machine Learning evaluation (precision/recall/F1), with a track record of taking projects from a trained model or rules engine to a live, publicly usable product.

## Technical Skills

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**Languages:** Java, JavaScript, TypeScript, SQL  
**Frontend:** React.js, Next.js, HTML5/CSS3, Tailwind CSS  
**Backend:** Spring Boot, Spring Security/JWT, Hibernate, REST APIs, Flask  
**AI / ML:** LLM Tool-Calling & Orchestration, Prompt Engineering, Model Evaluation, Scikit-learn, Random Forest, CNN/LSTM  
**Cloud & DevOps:** AWS (EC2, S3, Lambda, IAM, CloudWatch), Cloudflare Workers, Docker, Jenkins CI/CD, Git  
**Databases:** MySQL, PostgreSQL, Neo4j, JDBC  
**Data & Analytics:** Power BI, Tableau, Alteryx, Pandas  
**Core Concepts:** OOP, Data Structures & Algorithms, MVC Architecture, System Design

## Projects

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**ShipGuard AI :** *Spring Boot, Next.js, PostgreSQL, Neo4j, Qdrant, LLM Agents* | [Live Demo](#) | [Github](#)

- Built an AI-powered security-triage platform converting raw Semgrep/ESLint/gitleaks scanner output into a small, prioritized set of LLM-explained findings, cross-verified across models to cut false positives.
- Flagged recently-changed code as higher risk and modeled code dependencies as a Neo4j knowledge graph to visualize a change's blast radius, with Qdrant vector search surfacing semantically similar past findings.
- Deployed a multi-service architecture spanning Cloudflare Workers, Render, and managed PostgreSQL/Qdrant/Neo4j, with a Spring Boot API layer orchestrating the full LLM agent pipeline end to end.

**NRT AI :** *Next.js, TypeScript, Cloudflare Workers AI, LLM Tool-Calling* | [Live Demo](#) | [Github](#)

- Built a free, always-on AI assistant that chats, web-searches, generates/understands images, writes and runs code, and takes voice input, orchestrating multiple LLMs (Llama 3.3 70B, Moondream vision, Flux-1-schnell) behind a single interface.
- Designed a lightweight OpenAI-shape tool-calling router that judges when a query needs a tool call versus a direct model response, minimizing unnecessary or low-quality invocations.
- Implemented a resilient DuckDuckGo/Wikipedia search fallback chain and per-IP rate limiting via Cloudflare KV, shipped entirely on Workers AI's free tier with zero server billing or cold starts.

**LayoutIQ :** *Next.js, TypeScript, Cloudflare Workers, PDF Parsing* | [Live Demo](#) | [Github](#)

- Built a compliance tool for Licensed Town Planners that parses APDPMS-generated layout-plan PDFs (or manual entry) and validates plot dimensions, road widths, and open-space reservations against the selected authority's rules.
- Built a rules engine grading each submission against the exact governing regulation with a cited justification per violation and confidence flags for ambiguous parses, covering all AP DTCP and UDA authorities.
- Shipped a downloadable compliance report, fee estimator, and CSV export; deployed on Cloudflare Workers via [@opennextjs/cloudflare](#).

**Android Malware Detection System :** *Python, Flask, Scikit-learn, Random Forest* | [Live Demo](#) | [Github](#)

- Trained a Random Forest classifier on 10 key network-traffic features to classify Android traffic as benign or malicious in real time, catching malware that signature-based antivirus misses.
- Applied RobustScaler normalization to handle outlier-heavy traffic data and benchmarked precision, recall, and F1-score to validate accuracy before deployment.
- Deployed a Flask REST API (`/predict`, `/benign`, `/malware`) on Render with Gunicorn for live classification; extended the work into an IEEE-format paper presented at ICICST 2026.

### Web-Based Health Diagnosis System : *Java, JSP/Servlets, MySQL, Docker* | [Live Demo](#) | [Github](#)

- Built a full-stack health-diagnosis application in Java, JSP, and Servlets following MVC architecture, letting a clinician upload patient data and get an ML-assisted diagnosis instantly.
- Supported dataset upload and preprocessing, then compared CNN vs. LSTM performance on accuracy, precision, recall, and F1-score to pick the best-generalizing model for disease prediction.
- Migrated deployment to Docker on Render with a TiDB Cloud backend for scalable, production-style hosting.

### Water Quality Prediction System : *Python, Scikit-learn, Pandas* | [Live Demo](#) | [Github](#)

- Built an ML system predicting Ganga River water quality directly from physicochemical parameters (pH, dissolved oxygen, BOD, turbidity, conductivity), enabling faster, cheaper screening than lab testing.
- Engineered and selected features from raw sensor readings to feed the classification pipeline, handling noisy real-world environmental data.
- Trained and evaluated multiple Scikit-learn classification models with an 80/20 train-test split, benchmarking accuracy, precision, and recall to select the best-performing model.

## Experience

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### AWS Cloud Virtual Internship

Feb 2024 – May 2024

AICTE – Eduskills

Chennai, India

- Designed and deployed cloud-based applications using EC2, S3, and Lambda; configured IAM roles and least-privilege policies for secure access control.
- Monitored application performance and set up alerting with CloudWatch, and implemented cost-optimization strategies across compute and storage resources.

### Discipline Convenor

2022 – 2026

SRM University – Directorate of Student Affairs

Chennai, India

- Managed discipline and conduct processes for 500+ students, serving as the primary liaison between the student body and university administration on conduct-related matters.
- Coordinated cross-functional campus activities with faculty and administrative stakeholders, strengthening organizational, communication, and stakeholder-management skills.

## Certifications

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- Oracle Certified Java SE 11 Developer, Oracle | [Link](#)
- Oracle Certified Foundations Associate – Agentic AI, Oracle | [Link](#)
- ServiceNow Certified Application Developer (CAD), ServiceNow | [Link](#)
- ServiceNow Certified System Administrator (CSA), ServiceNow | [Link](#)
- Alteryx Designer Advanced Certification, Alteryx | [Link](#)

## Publications

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- “Android Malware Detection Using Static and Dynamic Analysis” – ICICST 2026, Dr. B.R. Ambedkar NIT Jalandhar, Apr 2026. IEEE-format paper on AI-driven mobile security combining static/dynamic analysis with ML.

## Education

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### SRM Institute of Science and Technology

Sept 2022 – May 2026

B.Tech in Computer Science and Engineering — CGPA: 8.64/10

Chennai, India

### Narayana IIT Academy

Apr 2020 – Jun 2022

Secondary Education (12th) — 68.8%

Nellore, India

### Sri Chaithanya Techno School

Jun 2019 – Jul 2020

10th Standard — 99%

Hyderabad, India